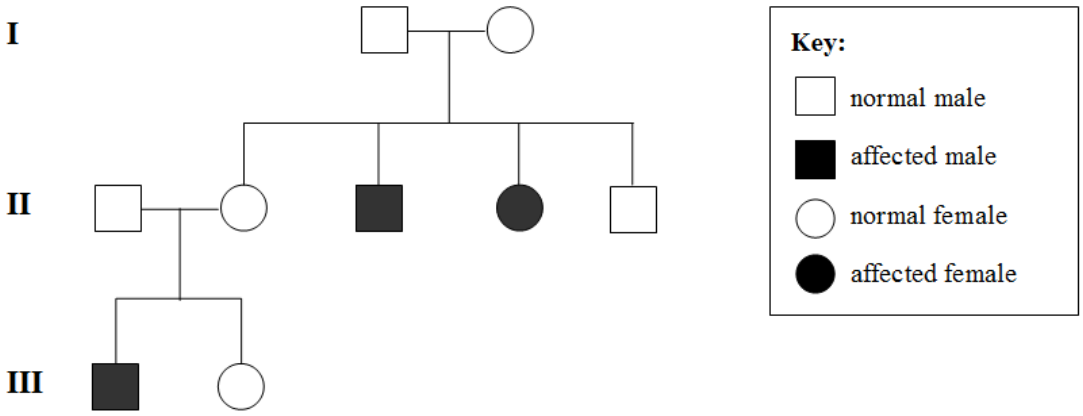
Multiple-choice questions

|!|BCEMB0260000001|!| ★

The pedigree below shows the inheritance of a genetic disease in a family.



Which of the following statements about the pedigree supports that the genetic disease is autosomal recessive but not autosomal dominant?

A. Both males and females can be affected.
B. Not all the offspring in generation II are affected.
C. Not all the offspring in generation III are affected.
D. The two pairs of normal parents have a child that has the genetic disease.

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0260000002|!| ★

In a DNA molecule of yeast, 15% of the nitrogenous bases are cytosine (C). What is the percentage of thymine (T) bases in this DNA molecule?

A. 15%
B. 30%
C. 35%
D. 70%

##C##

Skill assessed	Mark
Mathematical skills	1

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|!|BCEMC0260000003|!|

A woman with blood group AB has married a man with blood group O. What is the chance of their children having blood group B?

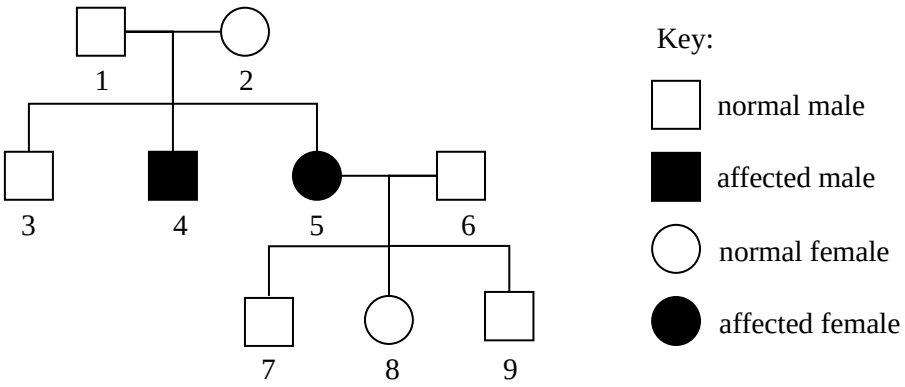
A. 0%
B. 25%
C. 50%
D. 75%

##C##

Skill assessed	Mark
Mathematical skills	1

|!|BCEMB0260000004|!| ★

Directions: The following three questions refer to the pedigree below, which shows the inheritance of a genetic disease in a family. The genetic disease is controlled by a pair of alleles.



Which of the following can be deduced from the pedigree above?

A. The disease-causing allele is dominant.
B. The disease-causing allele is recessive.
C. The disease-causing allele is sex-linked.
D. The genetic disease is caused by mutations in the pair of alleles in individuals 4 and 5.

##B ##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0260000005|!| ★

What are the probable genotypes of individuals 6 and 7?

	Individual 6	***Individual 7***
A.	homozygous	homozygous
B.	homozygous	homozygous or heterozygous
C.	homozygous or heterozygous	heterozygous
D.	homozygous or heterozygous	homozygous or heterozygous

☐

##C ##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0260000006|!| ★

If individuals 8 and 9 are twins, which of the following conclusions can be drawn?

(1) They must look alike.
(2) They developed from different fertilized eggs.
(3) They have the same DNA fingerprint.

A. (1) only
B. (2) only
C. (1) and (2) only
D. (1) and (3) only

☐

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0260000007|!|

DNA does ***not*** contain the nitrogenous base

A. adenine.
B. cytosine.
C. uracil.
D. guanine.

☐

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0260000008|!|

Which of the following statements about DNA is/are correct?

(1) Adenine is complementary to cytosine.
(2) DNA can be found in mitochondria.
(3) The whole length of a DNA molecule codes for proteins.

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0260000009|!|

Which of the following statements about chromosomes is ***incorrect***?

A. The number of chromosomes in a species is constant.
B. Chromosomes can only be observed during certain stages of the cell cycle.
C. Of the 23 pairs of chromosomes in humans, 22 pairs are autosomes.
D. The Y chromosome contains more genes than the X chromosome.

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0260000010|!|

Which of the following statements about DNA is/are correct?

(1) DNA is found in all cells in the human body.
(2) DNA has a double-helix structure.
(3) Complementary base pairs in a DNA molecule are held together by covalent bonds.

A. (2) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

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##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0260000011|!|

Which of the following elements is/are present in DNA?

(1) phosphorus
(2) nitrogen
(3) oxygen

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0260000012|!|

Human DNA can direct the synthesis of

(1) DNA polymerase.
(2) insulin.
(3) vitamin C.

A. (2) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMC0260000013|!|

The following shows some of the substances involved in heredity. Arrange these substances in ascending order of size.

(1) nucleotide
(2) thymine
(3) chromosome
(4) nucleus

A. (1) → (2) → (3) → (4)
B. (2) → (1) → (3) → (4)
C. (4) → (3) → (1) → (2)
D. (4) → (3) → (2) → (1)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0260000014|!|

In Mendel's breeding experiment to study the inheritance of pea plants, flowers that underwent cross-pollination were enclosed in a bag in order to

A. protect the flowers from mechanical damage.
B. increase the humidity around the enclosed flowers.
C. shield the flowers from sunlight.
D. prevent unwanted pollination.

##D##

Skill assessed	Mark
Practical skills	1

|!|BCEMC0260000015|!|

Which of the following statements about red-green colour blindness is ***incorrect***?

A. It is an X-linked trait.
B. It is more common in females than in males.
C. It is caused by a deficiency of red or green cone cells.
D. The allele for red-green colour blindness is recessive.

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0260000016|!|

Which of the following human characteristics shows discontinuous variation?

A. height
B. weight
C. sex
D. intelligence quotient (IQ)

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0260000017|!| ★

Which of the following statements about sex-linked genes in humans is correct?

A. Females can only be homozygous for the gene.
B. Males can only be heterozygous for the gene.
C. Females can be either heterozygous or homozygous for the gene.
D. Males can be either heterozygous or homozygous for the gene.

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0260000018|!| ★

A man of blood group AB marries a woman of blood group B. It is ***not*** possible for their children to have blood group

A. A.
B. B.
C. AB.
D. O.

##D##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0260000019|!| ★

In a family, the father and mother are of blood groups O and A respectively. They have a son who is of blood group A and a daughter who is of blood group O. Which of the following conclusions can be drawn?

(1) The mother is homozygous for blood group A.
(2) The son is heterozygous for blood group A.
(3) The next child of the couple has a 50% chance to be of blood group A.

A. (2) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0260000020|!|

Which of the following statements about human ABO blood groups is ***incorrect***?

A. Antigen A is present on the surface of red blood cells in people with blood group A.
B. Both antigens A and B are present on the surface of red blood cells in people with blood group AB.
C. People with blood group B produce anti-B antibodies.
D. People with blood group O produce anti-A and anti-B antibodies.

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0260000021|!|

Which of the following factors can lead to haemophilia?

(1) gene mutation
(2) inheritance of recessive alleles for haemophilia
(3) deficiency of calcium in the diet

A. (1) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0260000022|!| ★

Mrs Chan has blood group B. She and her husband gave birth to a son with blood group A and a daughter with blood group O. What is the chance of Mrs Chan and her husband giving birth to a daughter with blood group A?

A. 75%
B. 50%
C. 25%
D. 12.5%

##D##

Skill assessed	Mark
Mathematical skills	1

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|!|BCEMB0260000023|!| ★

People with blood group A can receive blood transfusions from people with blood group(s)

(1) A.
(2) AB.
(3) O.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0260000024|!| ★

Which of the following correctly shows the chromosome composition of an individual who has inherited an X chromosome from their father?

	Sex chromosomes	*Number of autosomes*
A.	XX	44
B.	XX	46
C.	XY	44
D.	XY	46

##A##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0260000025|!| ★

Directions: The following two questions refer to the table below, which shows the number of round seeds and wrinkled seeds produced by a pea plant that had been cross-pollinated.

Seed shape	Number of seeds
Round	28
Wrinkled	8

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Given that the allele for round seeds (R) is dominant to the allele for wrinkled seeds (r), what are the genotypes of the parent plants?

A. RR × RR
B. Rr × Rr
C. RR × Rr
D. Rr × rr

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0260000026|!| ★

What proportion of the seeds produced is homozygous for seed shape?

A. $^1/_4$
B. $^1/_2$
C. $^3/_4$
D. 0

##B##

Skill assessed	Mark
Mathematical skills	1

|!|BCEMB0260000027|!| ★

Directions: The following two questions refer to the information below.

In garden pea, the allele for tall stem (T) is dominant to the allele for short stem (t). A cross between a tall plant and a short plant gave rise to 106 tall plants and 94 short plants.

What is the probable genotype of the tall parent plant?

A. TT
B. Tt
C. tt
D. TT or Tt

##B##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0260000028|!| ★

Referring to the cross describe above, the actual result deviates from the predicted result based on Mendel's law of segregation. This is because

A. the genotype of the tall parent is not known.
B. the parent plants are not pure-breeding.
C. fertilization occurs at random.
D. mutation has occurred in the offspring.

##C##

Skill assessed	Mark
Critical thinking	1

|!|BCEMC0260000029|!|

Which of the following combinations correctly shows the type of variation that human height and blood groups exhibit?

	Height	***Blood group***
A.	continuous variation	continuous variation
B.	continuous variation	discontinuous variation
C.	discontinuous variation	continuous variation
D.	discontinuous variation	discontinuous variation

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0260000030|!|

Which of the following is ***least*** likely to cause differences in the appearance of identical twins after they have grown up?

A. differences in physical exercise
B. climate in the place they live in
C. nutritional differences
D. mutation

##D##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMC0260000031|!| **Cross topic**

Which of the following statements about DNA is ***incorrect?***

A. DNA contains ribose.
B. DNA contains thymine but not uracil.
C. There are about 10 base pairs per turn in a DNA double helix.
D. In a DNA molecule, bases on one strand are connected to complementary bases along the other strand by hydrogen bonds.

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0260000032|!| ★

In a family, the father and mother are of blood groups AB and B respectively. They have a son who is of blood group A. What is the probability that they will give birth to a daughter with blood group B?

A. $^3/_4$
B. $^1/_2$
C. $^1/_4$
D. $^1/_8$

##C##

Skill assessed	Mark
Mathematical skills	1

|!|BCEMC0260000033|!|

The term ‘alleles’ refers to

A. the locations of genes on a chromosome.
B. all of the genes on the same chromosome.
C. alternative forms of a gene.
D. different genes that determine the same characteristic.

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0260000034|!|

Directions: The following three questions refer to the information below.

In fruit flies, grey body is dominant to black body, and normal wings are dominant to vestigial wings.

If a heterozygous grey fruit fly is crossed with a black fruit fly, what percentage of the offspring would be black?

A. 25%
B. 50%
C. 75%
D. 100%

##B##

Skill assessed	Mark
Mathematical skills	1

|!|BCEMB0260000035|!| ★

A fruit fly having normal wings was crossed with a fruit fly having vestigial wings. All the F_1 offspring had normal wings. If these F_1 offspring were to cross with each other, what percentage of the F_2 offspring would have normal wings?

A. 25%.
B. 50%.
C. 75%.
D. 100%.

##C##

Skill assessed	Mark
Mathematical skills	1

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|!|BCEMA0260000036|!| ★★

Male fruit flies heterozygous for both grey body and normal wings were crossed with female fruit flies having black body and vestigial wings. The number of each type of offspring is shown in the table below.

Resulting offspring	Number
grey body, normal wings	952
black body, vestigial wings	948
grey body, vestigial wings	200
black body, normal wings	198

What conclusion can be drawn from the information given above?

A. The genes controlling body colour and wing shape are linked.
B. The genes are on different chromosomes.
C. The genes assort independently.
D. A mistake has been made.

□

##A##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0260000037|!| ★

In rabbit, long ears are dominant over short ears. If a rabbit heterozygous for long ears is mated with a short-eared rabbit, what would be the percentages of long-eared and short-eared offspring produced?

	Long-eared	***Short-eared***
A.	25%	75%
B.	50%	50%
C.	75%	25%
D.	100%	0%

□

##B##

Skill assessed	Mark
Mathematical skills	1

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|!|BCEMB0260000038|!| ★

Directions: The following two questions refer to the information below.

In pea plants, the flower colour is controlled by a pair of alleles. The allele for purple flowers (F) is dominant while the allele for white flower (f) is recessive. A pea plant having purple flowers was crossed with a pea plant with white flower. About half of the resulting F_1 plants had purple flowers and the other half had white flowers.

The genotypes of the parent plants were

A. FF and Ff.
B. Ff and Ff.
C. FF and ff.
D. Ff and ff.

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0260000039|!| ★

If the F_1 plants having purple flowers were allowed to self-pollinate, what percentage of the resulting F_2 plants would have purple flowers?

A. 25%
B. 50%
C. 75%
D. 100%

##C##

Skill assessed	Mark
Mathematical skills	1

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|!|BCEMC0260000040|!|

Which of the following is/are the component(s) of a chromosome?

(1) DNA
(2) RNA
(3) proteins

A. (1) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0260000041|!| ★

Gregor Mendel chose pea plants for his breeding experiments because

(1) they have observable contrasting traits.
(2) all the traits of pea plants are controlled by single genes.
(3) they can produce a large number of offspring.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Critical thinking	1

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|!|BCEMB0260000042|!| ★

In mice, the coat colour is determined by a pair of allele. A mouse homozygous for black coat was crossed with a white mouse, and all the F_1 mice produced were black. The F_1 mice were then allowed to interbreed. Which of the following statements about the F_2 generation are correct?

(1) 25% of the F_2 generation would be homozygous.
(2) Some of the black mice would be heterozygous.
(3) The expected ratio of black mice to white mice would be 3:1.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0260000043|!|

In pea plants, the petal colour is controlled by a pair of allele. A pea plant homozygous for purple flowers is crossed with a pea plant having white flowers, and the F_1 plants had purple flowers. One of these F_1 plants is then crossed with a pea plant having white flowers. What can be expected in the resulting offspring?

A. All plants having purple flowers
B. All plants having white flowers
C. 75% having purple flowers, 25% having white flowers
D. 50% having purple flowers, 50% having white flowers

##D##

Skill assessed	Mark
Analysing data	1

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|!|BCEMC0260000044|!|

The table below shows some characteristics of three newborn triplets.

	Sex	Weight (kg)	Body length (cm)	Blood group
Baby 1	Male	2	38	O
Baby 2	Male	2	40	A
Baby 3	Male	1.8	35	O

Which of the following conclusions can be drawn?

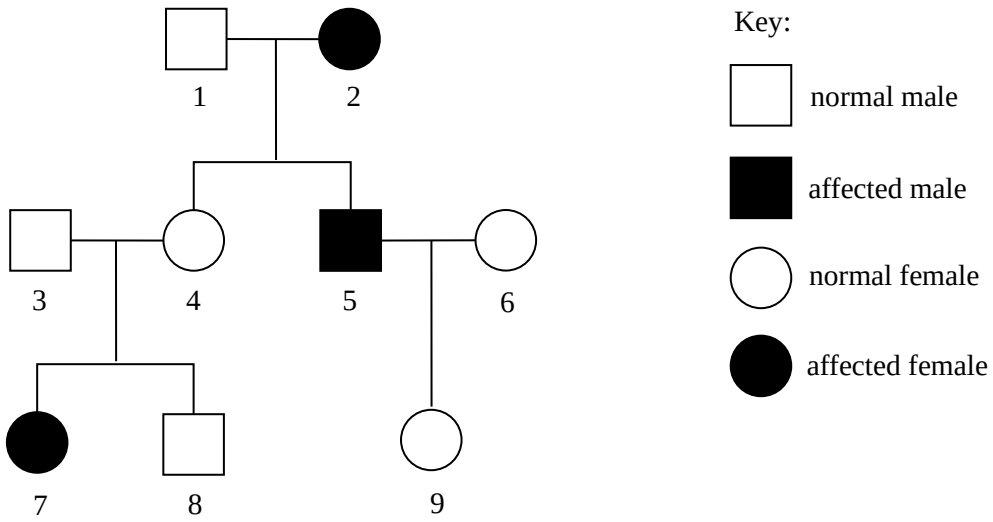
A. They are identical triplet.
B. Babies 1 and 2 are identical twins.
C. Babies 2 and 3 are non-identical twins.
D. They are non-identical triplets.

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0260000045|!| ★

The diagram below shows the inheritance of a disease caused by a recessive allele.



Which of the following individuals must be heterozygous?

A. 4 and 8
B. 4 and 9
C. 6 and 8
D. 6 and 9

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##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMC0260000046|!|

Which of the following statements about DNA replication is ***incorrect***?

A. It involves complementary base pairing.
B. DNA polymerase is required.
C. It occurs during interphase of the cell cycle.
D. It occurs in the cytoplasm of a eukaryotic cell.

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0260000047|!| ★

Directions: The following two questions refer to the information below.

In guinea pigs, the allele for short hair is dominant while the allele for long hair is recessive.

What will the F_1 offspring be from a cross between a heterozygous short-haired guinea pig and a long-haired guinea pig?

A. all short-haired
B. 25% short-haired and 75% long-haired
C. 50% short-haired and 50% long-haired
D. 75% short-haired and 25% long-haired

##C##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0260000048|!| ★

Two short-haired guinea pigs of the F_1 generation are crossed with each other. What will the F_2 offspring be?

A. all short-haired
B. all long-haired
C. 50% short-haired and 50% long-haired
D. 75% short-haired and 25% long-haired

□

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0260000049|!| ★

Which of the following statements about human sex chromosomes are correct?

(1) Sex chromosomes determine sex.
(2) The X chromosome is larger than the Y chromosome.
(3) Sex chromosomes carry some genes that may have nothing to do with sex.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

□

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0260000050|!|

Which of the following statements about human sex chromosomes are correct?

(1) Some genes on the X chromosome are absent from the Y chromosome.
(2) The X chromosome determines sex.
(3) A sperm carries either an X or a Y chromosome.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

□

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##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0260000051|!|

Which of the following is/are the cause(s) of genetic variations?

(1) mutation
(2) independent assortment of chromosomes at meiosis
(3) random fertilization

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0260000052|!|

Which of the following is ***not*** an example of continuous variation?

A. height
B. weight
C. blood groups
D. intelligence quotient

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0260000053|!|

Which of the following is an example of continuous variation?

A. blood groups
B. eye colour
C. the shape of ear lobes
D. hand span

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##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0260000054|!| ★

Albinism (a condition in which pigmentation is lacking) in humans is caused by a recessive allele. Two parents with normal skin pigment have an albino child. If they are going to have another child, what would be the probability of this child having normal skin pigment?

A. 100%
B. 75%
C. 50%
D. 33.3%

##B##

Skill assessed	Mark
Mathematical skills	1

|!|BCEMC0260000055|!|

Which of the following is a sex-linked trait?

A. red-green colour blindness
B. blood group
C. albinism
D. height

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0260000056|!| ★

Directions: The following three questions refer to the information below.

In fruit flies, the allele for long wings (L) is dominant to the allele for vestigial wings (l). In an investigation, long-winged male fruit flies were mated with long-winged female fruit flies. The numbers of different types of offspring were counted. The results are shown in the table below.

	Male flies	**Female flies**	**Total**
Long-winged flies	51	54	105
Vestigial-winged flies	19	20	39
Total	70	74	144

What are the genotypes of the parents?

A. LL × LL
B. Ll × Ll
C. LL × Ll
D. Ll × ll

□

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0260000057|!| ★

Which of the following determine(s) the sex ratio in the offspring produced?

(1) random fertilization
(2) crossing over
(3) mutation

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

□

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0260000058|!| ★

What proportion of the long-winged offspring is homozygous?

A. $^1/_4$

B. $^1/_3$

C. $^1/_2$

D. $^3/_4$

##B##

Skill assessed	Mark
Mathematical skills	1

|!|BCEMB0260000059|!| ★

In fruit flies, long wings are dominant to vestigial wings. Which of the following crosses will produce long-winged and vestigial-winged offspring in approximately a 1:1 ratio?

A. heterozygous long-winged × heterozygous long-winged

B. heterozygous long-winged × homozygous long-winged

C. heterozygous long-winged × vestigial-winged

D. homozygous long-winged × vestigial-winged

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0260000060|!|

Which of the following factors can determine the skin colour of a person?

(1) the skin colour of the person's parents

(2) exposure to sunlight

(3) vitamin D consumption

A. (2) only

B. (1) and (2) only

C. (2) and (3) only

D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0260000061|!|

Which of the following comparisons between characteristics showing continuous and discontinuous variations is ***incorrect***?

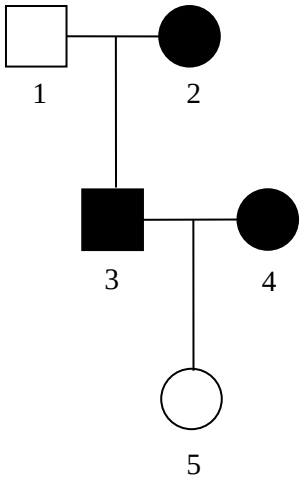
	Continuous variation	***Discontinuous variation***
A.	usually controlled by many genes	usually controlled by one gene
B.	show normal distribution	do not show normal distribution
C.	may be influenced by environmental factors	less likely to be influenced by environmental factors
D.	show no intermediate phenotypes	show intermediate phenotypes

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0260000062|!| ★

Directions: The following two questions refer to the pedigree below, which shows the inheritance of the shape of earlobes in a family. The shape of a person's earlobes is determined by a pair of alleles. The allele for free earlobes (E) is dominant to the allele for attached earlobes (e).



Key:


Male having attached earlobes


Male having free earlobes


Female having attached earlobes


Female having free earlobes

What are the genotypes of individuals 1 and 3?

	Individual 1	***Individual 3***
A.	ee	EE
B.	ee	Ee
C.	EE	Ee
D.	Ee	Ee

##B##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0260000063|!| ★

If individuals 3 and 4 are going to have a next child, what is the probability of this child having free earlobes?

A. 0%
B. 25%
C. 50%
D. 75%

##D ##

Skill assessed	Mark
Mathematical skills	1

|!|BCEMB0260000064|!| ★

In humans, curly hair is dominant to straight hair and free earlobes are dominant to attached earlobes. A man and a woman are heterozygous for both characteristics. What is the probability that their child will have straight hair and attached earlobes?

A. 0
B. $^{1}/_{16}$
C. $^{3}/_{16}$
D. $^{9}/_{16}$

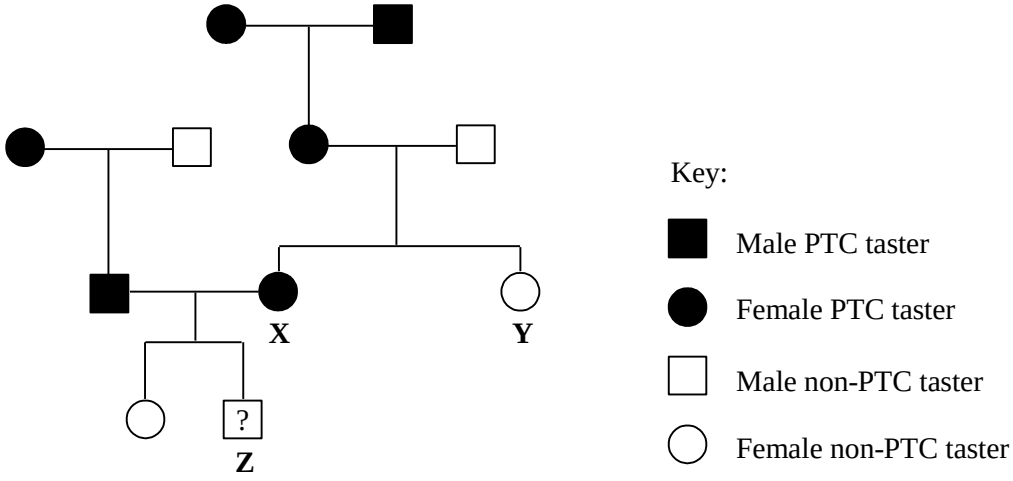
##B##

Skill assessed	Mark
Mathematical skills	1

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|!|BCEMB0260000065|!| ★

Directions: The following two questions refer to the pedigree below, which shows the inheritance of the ability to taste bitterness from the compound phenylthiocarbamide (PTC). Only people who have the dominant allele, T, are tasters of PTC.



What are the genotypes of individuals X and Y?

	Individual X	*Individual Y*
A.	TT	tt
B.	Tt	Tt
C.	Tt	tt
D.	tt	Tt

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0260000066|!| ★

What is the probability that individual Z is a PTC taster?

A. 25%
B. 50%
C. 75%
D. 100%

##C##

Skill assessed	Mark
Mathematical skills	1

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|!|BCEMC0260000067|!|

Which of the following trait is ***not*** entirely controlled by genes?

A. intelligence
B. sex
C. blood group
D. eye colour

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0260000068|!|

Some genetic diseases are not easy to diagnose in childhood because the symptoms are not yet present.

Which of the following methods can help doctors predict whether a child will develop genetic diseases later in life?

(1) analysis of the child's DNA sequence
(2) analysis of the child's karyotype
(3) pedigree analysis

A. (1) only
B. (3) only
C. (1) and (2) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0260000069|!| ★

A colour-blind woman, Amy, married a man with normal colour vision. They have a daughter and a son. Which of the following conclusions can be made?

(1) Amy's father is colour-blind.
(2) Amy's mother is colour-blind.
(3) Amy's daughter carries an allele for colour blindness.

A. (3) only
B. (1) and (2) only
C. (1) and (3) only
D. (1), (2) and (3)

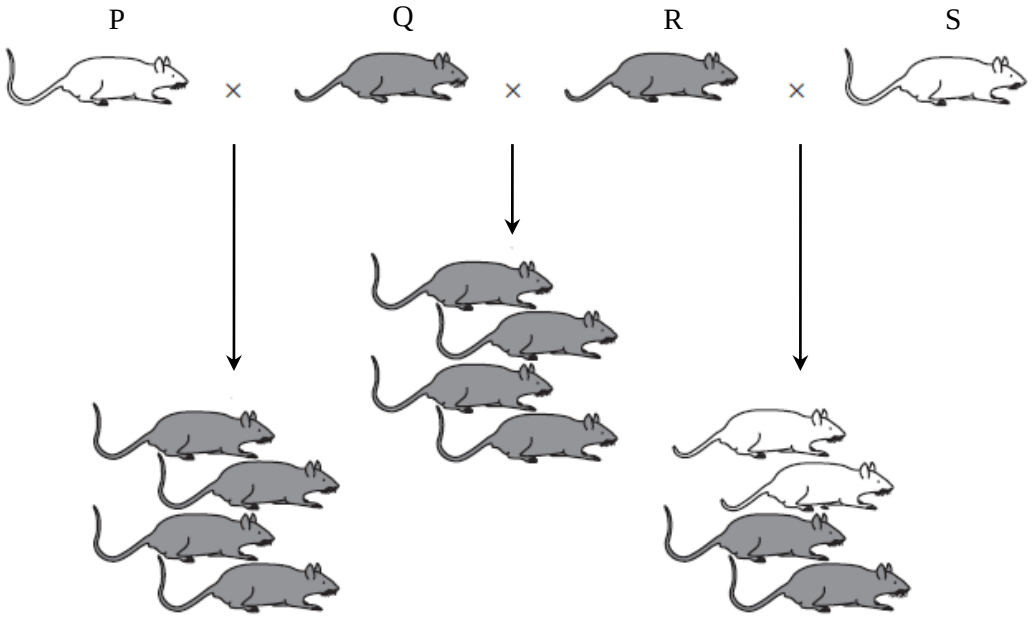
□

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0260000070|!| ★

In mice, the coat colour is controlled by a pair of alleles. The allele for black coat is dominant to the allele for white coat. The diagram below shows the results of three crosses of mice.



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Which of the mice, P, Q, R or S, is heterozygous for coat colour?

A. P
B. Q
C. R
D. S

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0260000071|!| ★

In guinea pigs, black hair is dominant to brown hair. What phenotypic ratio would be expected from a cross between a heterozygous black guinea pig and a brown guinea pig?

A. all black
B. 1 black : 1 brown
C. 1 black : 3 brown
D. 3 black : 1 brown

##B##

Skill assessed	Mark
Analysing data	1

|!|BCEMA0260000072|!| ★★

The fur colour of guinea pigs is controlled by a pair of alleles. Two black guinea pigs were mated and produced 9 black and 3 white guinea pigs. Which of the following conclusions can be drawn?

(1) White fur colour was the result of a mutation.
(2) Both parents were heterozygous for black fur.
(3) The black guinea pigs in the F_1 generation can be homozygous or heterozygous.

A. (2) only
B. (3) only
C. (1) and (3) only
D. (2) and (3) only

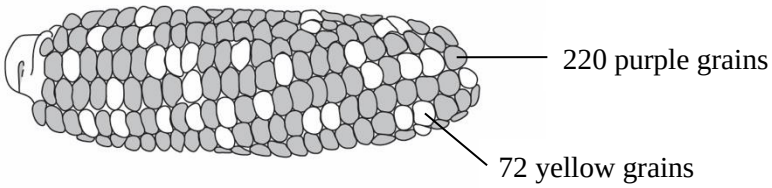
##D##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0260000073|!| ★

In maize, the grain colour is controlled by a pair of alleles. The allele for purple grains (G) is dominant while the allele for yellow grains (g) is recessive. The diagram below shows a maize cob with purple and yellow grains. The two types of grains have been counted.



What are the genotypes of the parent plants of this maize cob?

A. GG × gg
B. Gg × gg
C. Gg × Gg
D. gg × gg

□

##C ##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0260000074|!| ★

In fruit flies, the body colour is controlled by a pair of alleles. The allele for grey body is dominant while the allele for black body is recessive. Fruit flies which are heterozygous for body colour are allowed to breed and 200 offspring are produced. How many of the offspring are expected to be black-bodied?

A. 0
B. 50
C. 100
D. 150

□

##B##

Skill assessed	Mark
Mathematical skills	1

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|!|BCEMB0260000075!| ★

In a certain kind of cats, folded ears is a trait controlled by a pair of allele. A cross between two cats with folded ears produced four kittens, of which three had straight ears and one had folded ears. Which of the following is correct?

	Conclusion	***Evidence***
A.	Folded ear is dominant.	The phenotypic ratio of the offspring is 3:1.
B.	Folded ear is dominant.	Two parents with folded ears produced offspring with straight ears.
C.	Straight ear is dominant.	The phenotypic ratio of the offspring is 3:1.
D.	Straight ear is dominant.	Two parents with folded ears produced offspring with straight ears.

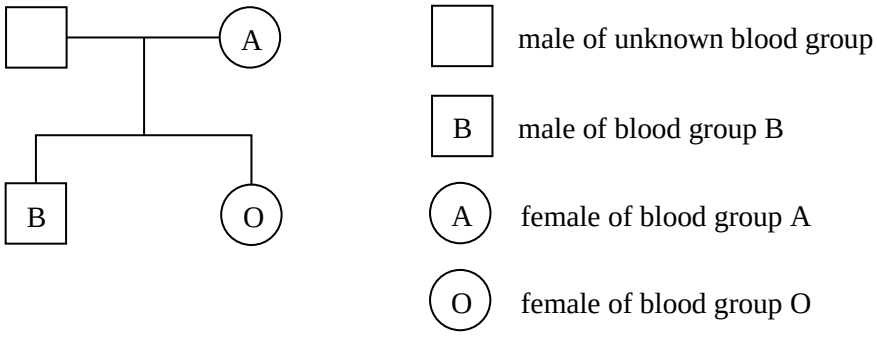
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##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0260000076|!| ★

The pedigree below shows blood groups of a mother and two children.



What is the blood group genotype of the father?

A. ii
B. I^AI^B
C. I^Bi
D. I^BI^B

□

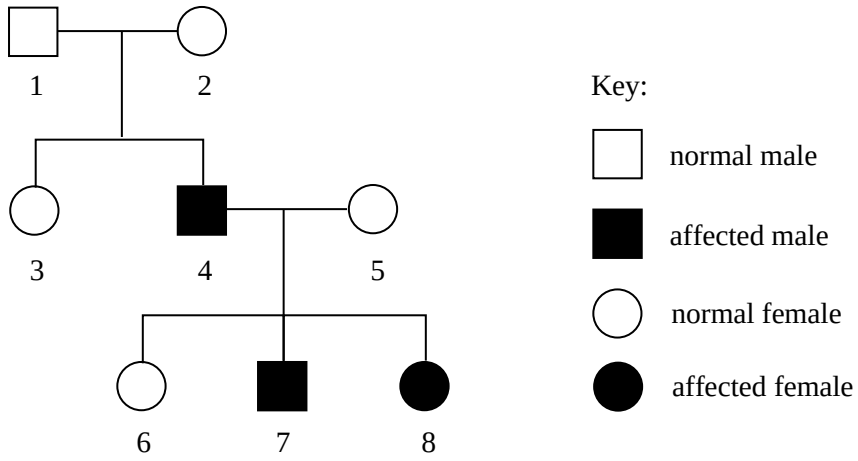
##C##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0260000077|!| ★

The pedigree below shows the inheritance of a disease in a family. The disease is caused by a recessive allele on the X chromosome.



Which of the following individuals must be carriers of the recessive allele?

A. individuals 2 and 5 only
B. individuals 2 and 6 only
C. individuals 2, 5 and 6 only
D. individuals 2, 3, 5, and 6

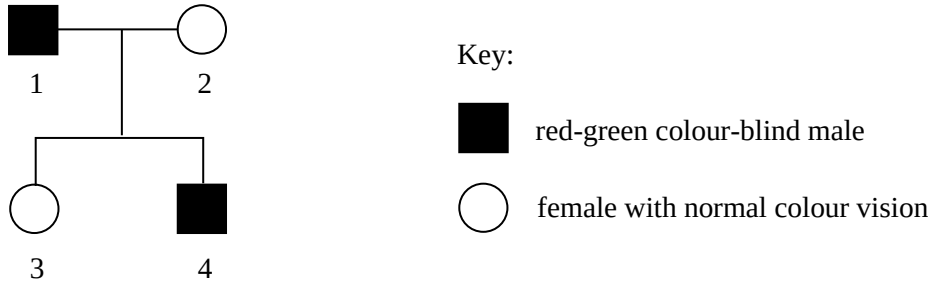
□

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0260000078|!| ★

The pedigree below shows the inheritance of red-green colour blindness in a family.



Which of the following can be concluded from the pedigree?

(1) Individual 1 has two alleles for red-green colour blindness.
(2) Individual 2 is a carrier.
(3) Individual 3 is a carrier.

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A. (1) only
B. (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0260000079|!| ★

In a certain kind of cats, the fur length is controlled by a pair of alleles. The table below shows three crosses.

	Number of offspring	
	long fur	**short fur**
Cross 1: long fur × long fur	26	13
Cross 2: short fur × short fur	0	38
Cross 3: long fur × short fur	21	23

Which of the above crosses can be used to deduce which allele for the fur length is dominant?

A. cross 1 only
B. cross 2 only
C. cross 3 only
D. cross 2 and 3 only

##A##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0260000080|!| ★

Which of the following are the possible blood group genotypes of a couple who have a child of blood group AB?

(1) $I^AI^B \times I^AI^B$
(2) $I^AI^B \times ii$
(3) $I^Ai \times I^Bi$

A. (1) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0260000081|!|

Which of the following statements about a red-green colour-blind female must be correct?

A. Both of her parents are red-green colour-blind.
B. Her mother is a carrier of red-green colour blindness.
C. Her daughter is red-green colour-blind.
D. Her son is red-green colour-blind.

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0260000082|!|

Which of the following is ***not*** a genetic disease in humans?

A. haemophilia
B. albinism
C. sickle cell anaemia
D. scurvy

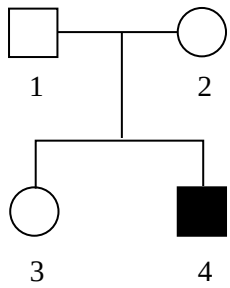
##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMA0260000083|!| ★★

Directions: The following two questions refer to the pedigree below, which shows the inheritance of red-green colour blindness in a family.



Key:

female with normal colour vision

male with normal colour vision

red-green colour-blind male

Let B represent the allele for normal colour vision and b represent the allele for red-green colour blindness. Which are the possible genotypes of individuals 2 and 3?

	Individual 2	***Individual 3***
A.	X^BX^B	X^BX^B and X^BX^b
B.	X^BX^B	X^BX^B
C.	X^BX^b	X^BX^B and X^BX^b
D.	X^BX^b	X^BX^B

□

##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0260000084|!| ★

What is the probability of the couple's next son being red-green colour-blind?

A. 0%
B. 50%
C. 75%
D. 100%

□

##B##

Skill assessed	Mark
Mathematical skills	1

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|!|BCEMB0260000085|!| ★

In pea plants, seed shape and seed colour are controlled by two genes located on different chromosomes. The allele for round seeds (R) is dominant while the allele for wrinkled seeds (r) is recessive. The allele for yellow seeds (Y) is dominant while the allele for green seeds (y) is recessive.

If pea plants having genotypes RRYy and rryy are crossed, the possible genotypes of the offspring would be

A. RRYy and RRyy.
B. RRyy and rrYy.
C. RrYy and Rryy.
D. RrYy and rryy.

☐

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0260000086|!|

Which term describes the alternative forms of a hereditary factor that controls a particular characteristic?

A. alleles
B. genotypes
C. genes
D. chromosomes

☐

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0260000087|!| ★

Red-green colour-blindness is an X-linked trait. The allele for normal colour vision (B) is dominant and the allele for red-green colour blindness (b) is recessive.

A woman who is a carrier of red-green colour blindness marries a man who has normal colour vision. Which of the following is ***not*** a possible genotype of their children?

A. X^BX^b
B. X^bX^b
C. X^BY
D. X^bY

☐

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##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0260000088|!| ★

The ABO blood groups in humans are determined by multiple alleles, I^A, I^B and i. I^A represents the allele for blood group A, I^B represents the allele for blood group B and i represents the allele for blood group O.

The blood groups of a couple are AB and O. What is/are the possible genotype(s) of their children?

A. I^AI^B only
B. I^Ai and ii
C. I^AI^B and ii
D. I^Ai and I^Bi

□

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0260000089|!|

Which of the following statements best describes Mendel's first law?

A. Alleles separate so that half of the gametes receive one allele and the other half of the gametes receives the other allele.
B. The traits of organisms are controlled by one or several pairs of alleles.
C. In the heterozygous condition, the dominant allele can express its effect.
D. The separation of alleles for one gene is independent of the separation of alleles for other genes during gamete formation.

□

##A##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMA0260000090|!| ★★

In rabbits, the allele for long ears (L) is dominant while the allele for short ears (l) is recessive. The allele for black fur (B) is dominant while the allele for brown fur (b) is recessive. It is known that the genes controlling the ear length and the fur colour are located on different chromosomes. If a rabbit heterozygous for both characteristics are crossed with a short-eared brown rabbit, what proportion of the offspring would be expected to be short-eared brown rabbit?

A. $^{1}/_{16}$

B. $^{1}/_{4}$

C. $^{3}/_{16}$

D. $^{9}/_{16}$

□

##B##

Skill assessed	Mark
Mathematical skills	1

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